«CUSTOMER NAME»
«GROUP COMPANIES»

Predictive Analytics · Service B

Production Deployment of Predictive Models

Technical and Commercial Proposal

Scheduled scoring and forecasts, delivery to business systems, monitoring and handover on a shared MLOps foundation

Proposal No: KOMTAS-«CUSTOMER ABBREVIATION»-PRA-B-«YEAR»-«SEQUENCE»

Proposal Date: «dd Month yyyy»

Validity: «proposal date + 1 month»

VARIABLE GLOSSARY — in this template only the fields inside « » change per customer. All other text is fixed.

Identity: «CUSTOMER» short customer name used in the text · «CUSTOMER NAME» full legal name (cover) · «CUSTOMER ABBREVIATION» short code in the proposal number · «YEAR» year · «SEQUENCE» proposal sequence number · «dd Month yyyy» proposal date · «proposal date + 1 month» end of validity · «price validity date» latest start date for which the fees stay valid · «GROUP COMPANIES» group companies in scope; delete the line for a single company

Context: «CUSTOMER CONTEXT» one paragraph: company, sector, data platform, business goal of the models, result of the Pilot · «PILOT REF» number and date of the Pilot (service A) report, or of the document describing the equivalent validated model, on which this proposal is based

Scope: «BAND» sizing band S, M or L from the proposal input form · «TARGET SYSTEM» business system that receives the scores, forecasts or recommendations (e.g. CRM, campaign tool, ERP / replenishment, e-commerce site) · «FREQUENCY» scoring or forecast frequency (daily, weekly, monthly) · «HORIZON» forecast horizon (e.g. 1 to 12 weeks ahead) · «PLATFORM» delivery platform agreed in the Pilot: Google Cloud, Dataiku, Azure Machine Learning or an open-source stack · «USE CASE» use case from the catalogue in Annex 1, e.g. Path to Churn, demand forecasting, collection optimization

Families: «FAMILY — RETENTION & CHURN» text that applies only if a use case of the Retention & Churn family is in scope · «FAMILY — DEMAND & SUPPLY» text that applies only if a use case of the Demand & Supply family is in scope · «FAMILY — PRICING & PERSONALIZATION» text that applies only if a use case of the Pricing & Personalization family is in scope · «FAMILY — RISK & COLLECTIONS» text that applies only if a use case of the Risk & Collections family is in scope · «FAMILY — INDUSTRIAL OPERATIONS» text that applies only if a use case of the Industrial Operations family is in scope

Values: «NUMBER» numeric value (count, threshold, days, weeks, person-days) · «%» percentage · «RESULT» measured result approved for use by the reference customer; never estimated

Commercial: «fee» fee excl. VAT · «amount» payment amount excl. VAT · «day rate» person-day rate excl. VAT (rate card) · «CITY» city where on-site meetings are free of charge

Contacts: «full name» contact name · «title» contact title · «e-mail / phone» contact details

Markers: the five family blocks are resolved as described in the first fill note; «PLATFORM» is the delivery platform agreed in the Pilot. The orange AX DOĞRULASIN and GÜNER DOĞRULASIN tags mark draft statements to be verified before use. FILL NOTE boxes are internal and are deleted before sending.

Contents

FILL NOTE — this template is used only after service A (Pilot) or for a model of «CUSTOMER» that has been validated in an equivalent way. The Pilot report or the model document is referenced as «PILOT REF». If neither exists, this template is not used; the Pilot is proposed first.

Use-case families: every heading, paragraph, bullet, table row and annex page starting with «FAMILY — RETENTION & CHURN», «FAMILY — DEMAND & SUPPLY», «FAMILY — PRICING & PERSONALIZATION», «FAMILY — RISK & COLLECTIONS» or «FAMILY — INDUSTRIAL OPERATIONS» is kept only for the families of the use cases in this proposal; the others are deleted. In the parts kept, only the marker is deleted. Where two use cases of one family are in scope, the rows of that family are repeated per use case. The MLOps foundation is counted once, whatever the number of use cases.

«PLATFORM» is the delivery platform agreed in the Pilot (Google Cloud, Dataiku, Azure Machine Learning or an open-source stack). In Table 6 only the row of that platform is kept.

Integrations: one row per target system in Table 4 and one fee line per integration in Table 16.

Statements followed by an orange tag are draft assumptions of the package. AX DOĞRULASIN: use cases, models, data requirements, metrics, effort, durations, team, references, results and acceptance, verified with the Komtaş Analytics team. GÜNER DOĞRULASIN: cloud platform items and two references, verified with Cloud Solutions. After verification only the tag is deleted. No proposal is sent with a tag in it.

Order of completion: «PILOT REF» → use cases (Section 1) → «PLATFORM» → Table 2 → integrations (Table 4) → effort and durations (Table 9) → team → fees (Section 9) → Annex 1 → Annex 2 → remaining « » fields. Result figures of earlier projects are written only as given in Table 25 or as approved by the customer concerned. Margin, discount or consumption figures are never written into this proposal.

1. Purpose

«CUSTOMER CONTEXT»

The aim of this proposal is to take the predictive models validated in the Pilot «PILOT REF» into production on «PLATFORM», the delivery platform of «CUSTOMER». In production the models are retrained on a schedule, produce scores or forecasts at the agreed frequency, deliver them to the business systems that use them and are monitored for drift and performance.

The unit of the work is the use case. The use cases in scope of this proposal and their families from the catalogue in Annex 1 are:

- **«FAMILY — RETENTION & CHURN» Retention & Churn — «USE CASE»:** customers likely to leave, cancel a subscription or move to a lower segment, with the main drivers, for retention actions.
- **«FAMILY — DEMAND & SUPPLY» Demand & Supply — «USE CASE»:** demand, sales or visitor forecasts per product, store, site or channel and horizon, for planning, replenishment and stock.
- **«FAMILY — PRICING & PERSONALIZATION» Pricing & Personalization — «USE CASE»:** offers, recommendations, segments, customer value or prices per customer, product or segment, for campaigns and pricing decisions.
- **«FAMILY — RISK & COLLECTIONS» Risk & Collections — «USE CASE»:** probability of delinquency or default per customer, with the main drivers, for collection contact policy, credit and limit decisions.
- **«FAMILY — INDUSTRIAL OPERATIONS» Industrial Operations — «USE CASE»:** predicted process values (energy, temperature, quality, yield, equipment condition) and recommended set points, for production and maintenance teams.

A shared MLOps foundation is built once and used by all use cases. The fee consists of one line for the MLOps foundation, one line per use case and one line per integration with a target system (Section 9).

2. Scope

2.1 Service Structure

Predictive Analytics consists of three services; this proposal is for service B.

Service	Focus	Result
A — Pilot	One use case: data readiness check, feature set, model, validation and first business value measurement in 6 to 8 weeks «AX DOĞRULASIN»	Fixed fee; go / no-go decision for Production
B — Production	MLOps foundation, deployment of the use cases, scheduled scoring, integration, monitoring, handover	Models running in production; fee per use case
C — Run	Monthly monitoring, retraining, reporting and small improvements	Separate proposal (Section 10)

Table 1 — Service structure. B is offered only on the basis of a validated model.

2.2 Prerequisite

This proposal is based on the Pilot «PILOT REF». The data readiness report, the feature definitions, the validation report and the value measurement plan of the Pilot are the basis of the work.

If a use case has not gone through a Komtaş Pilot but «CUSTOMER» has an equivalent validated model, the model is reviewed in Phase 0 (data, features, validation on hold-out data). Production work on that use case starts only after the review is accepted. «AX DOĞRULASIN»

2.3 Sizing Basis

Sizing is based on the values below, taken from the Pilot «PILOT REF» and the proposal input form. The values are confirmed in Phase 0 (Section 7.1).

Sizing Input	Value
Use cases in scope	«NUMBER»
Delivery platform	«PLATFORM»
Data sources used by the models	«NUMBER»
«FAMILY — RETENTION & CHURN» Customers scored	«NUMBER»
«FAMILY — DEMAND & SUPPLY» Time series forecast (product × store, site or channel)	«NUMBER»
«FAMILY — DEMAND & SUPPLY» Forecast horizon	«HORIZON»
«FAMILY — PRICING & PERSONALIZATION» Customers and products or SKUs	«NUMBER»
«FAMILY — RISK & COLLECTIONS» Customers or credit applications scored	«NUMBER»
«FAMILY — INDUSTRIAL OPERATIONS» Production units and sensor tags	«NUMBER»
Prediction frequency	«FREQUENCY»
Target systems to integrate	«NUMBER»
Sizing band	«BAND»

Table 2 — Values on which sizing is based. Source: «PILOT REF» and the proposal input form.

The band (S / M / L) sets the effort per use case. It is the highest band reached by at least three of the drivers data sources, volume, prediction frequency, target systems and number of use cases. «AX DOĞRULASIN»

2.4 Production Scope per Family

The production scope of each use case follows its family: «AX DOĞRULASIN»

Family	Production Scope	Output Delivered
«FAMILY — RETENTION & CHURN» Retention & Churn	Retraining on the full history, scheduled scoring of all active customers, driver explanation per customer, monitoring	Churn or downgrade probability and main drivers per customer, refreshed «FREQUENCY»
«FAMILY — DEMAND & SUPPLY» Demand & Supply	Retraining, scheduled forecast for all series in scope, back-test against the current method, monitoring	Forecast per series for the horizon «HORIZON», refreshed «FREQUENCY»
«FAMILY — PRICING & PERSONALIZATION» Pricing & Personalization	Retraining, scheduled scoring or price calculation, business rules (stock, exclusions, price limits), monitoring	Offer, recommendation list, segment, customer value or price per customer, product or segment, refreshed «FREQUENCY»
«FAMILY — RISK & COLLECTIONS» Risk & Collections	Retraining, scheduled scoring of the portfolio, explanation of the drivers (feature importance and direction of impact), monitoring	Delinquency or default probability and main drivers per customer, input to contact policy, credit or limit decisions, refreshed «FREQUENCY»
«FAMILY — INDUSTRIAL OPERATIONS» Industrial Operations	Retraining on process, sensor and laboratory data, scheduled prediction, set point recommendation within operating limits, monitoring	Predicted process values, quality class or recommended set points per unit, refreshed «FREQUENCY»

Table 3 — Production scope and outputs per family.

2.5 Integration with Target Systems

The outputs are delivered to the business systems below, for example CRM, campaign tool, pricing or collection system, ERP or replenishment, or the operator screens of a plant. Each integration is a separate line in the fee (Section 9).

#	Target System	Use Case	Method	Frequency
I1	«TARGET SYSTEM»	«USE CASE»	Output table on «PLATFORM» read by the target system, file export or API call «AX DOĞRULASIN»	«FREQUENCY»
I2	«TARGET SYSTEM»	«USE CASE»	Output table, file export or API call	«FREQUENCY»

Table 4 — Integrations. One row per target system; the use case column names the use case whose output is delivered.

Komtaş delivers the outputs up to the interface of the target system. Changes inside the target system (screens, campaign rules, pricing or replenishment logic, process control systems) are made by «CUSTOMER» or its vendor. «AX DOĞRULASIN»

2.6 Boundaries to Other Services

- **AI Readiness Assessment:** measures the readiness of the organisation for AI; this package builds and operates the models.
- **MLOps / AI Factory platform:** an enterprise platform for many models and teams; separate service. This package builds the MLOps foundation needed by the use cases in scope.
- **Analytic CoE Enablement Program:** builds the analytics team, roadmap and governance of the organisation; separate service.
- **Analytics platform migration:** moves existing models and data flows to a new analytics platform (for example from SAS, R or Pentaho to Dataiku); separate service.
- **Data platform modernisation:** moves the data platform (for example on-premises ETL to Google Cloud); separate services.
- **Gemini Agent Kit:** covers generative AI agents and retrieval-augmented generation (RAG); separate service.
- **AI Governance Readiness:** covers the governance of models and AI systems; separate service.

3. Approach

3.1 Target Architecture

The models run in the layers below. The layers, the lifecycle and the phases are the same on every delivery platform; the platform sets the components (Section 3.2).

Layer	Role
Data	Source data loaded from the source systems, sensors or files; feature tables
Training	Training and retraining of the models
Model registry	Versions of the models, approved model in production
Pipelines and scheduling	Scheduled feature preparation, training and scoring; notification of failed runs
Prediction	Scores, forecasts, recommendations or set points at the agreed frequency (batch)
Output	Output tables as single source for the target systems and reports
Delivery	Export to the target systems (Table 4)
Monitoring and reporting	Data drift, prediction drift and model performance against thresholds; alerts; report of model performance and business metrics

Table 5 — Target architecture layers.

3.2 Delivery Platform

The work is delivered on «PLATFORM». Komtaş delivers models on the platforms below; the platform is agreed in the Pilot and is part of the scope. «AX DOĞRULASIN»

Platform	Components Used
Google Cloud	BigQuery for data, feature and output tables; BigQuery ML or Vertex AI for training; Vertex AI Model Registry; Vertex AI Pipelines and batch prediction; Looker for reporting «GÜNER DOĞRULASIN»
Dataiku	Flows for feature preparation and training; scenarios for scheduled retraining and scoring; automation node for production runs; data and model stability monitoring with conditional and periodic retraining
Azure Machine Learning	Azure Machine Learning for training, model registry, pipelines and batch scoring; Azure Data Factory or Microsoft Fabric for data movement; Azure Synapse or a Fabric warehouse for data and output tables
Open-source stack	Open-source components for pipelines, model registry, scheduling and monitoring on the infrastructure of «CUSTOMER»; components agreed in Phase 0

Table 6 — Delivery platform options. Only the row of «PLATFORM» is kept.

3.3 MLOps Foundation

The MLOps foundation is built once in Phase 1 and used by all use cases, including use cases added later: «AX DOĞRULASIN»

- code repository and CI/CD for the pipelines;
- pipeline template for feature preparation, training, validation and scoring;
- model registry with versioning and an approval step before a model goes into production;
- scheduling of the runs and notification of failed runs;
- monitoring of drift and performance with thresholds and alerts;
- separate development and production environments, access and service accounts.

Komtaş has built MLOps foundations for industrial and banking customers on Google Cloud with Dataiku, on Dataiku and on open-source stacks (Table 26, Annex 2).

3.4 Production Steps per Use Case

Each use case goes through the steps below. «AX DOĞRULASIN»

Step	What Is Done
Entry	Pilot model, feature definitions and validation result confirmed; data feeds for production checked
Feature pipeline	Features of the Pilot rebuilt as a scheduled pipeline on the production data
Retraining and validation	Model retrained on the full history and validated on hold-out data against the Pilot result
Scoring	Scheduled scoring or forecast at the agreed frequency; output tables on «PLATFORM»
Delivery	Outputs delivered to the target systems (Table 4)
Monitoring	Drift and performance thresholds set; alerts and monitoring report in place
Parallel run and go-live	Scheduled runs checked together with the business owner; go-live approval

Table 7 — Production steps per use case.

3.5 Measurement of Business Value

The control group or hold-out design agreed in the Pilot is continued in production. The business metric of each use case is measured over «NUMBER» weeks after go-live and reported. «AX DOĞRULASIN»

Business value can only be measured if «CUSTOMER» acts on the outputs (for example runs retention campaigns on the scored customers or applies the recommended prices) and keeps the control group. The business result is measured; it is not guaranteed. The results in Table 25 were measured in the same way, as test group against control group.

4. Activities and Timeline

4.1 Phases and Exit Criteria

Each phase has a written exit criterion; the next phase does not start until it is met. Phases 2 to 4 are run per use case. «AX DOĞRULASIN»

Phase	What Is Done	Exit Criterion
0 — Kick-off and entry	Plan, access, confirmation of the sizing values, platform, target systems and frequency; review of an equivalent model where needed	Plan and sizing values approved in writing
1 — MLOps foundation	Repository, CI/CD, pipeline template, model registry, scheduling, monitoring, environments (Section 3.3)	Pipeline template runs end to end in the production environment
2 — Use case in production	Feature pipeline, retraining, validation, scheduled scoring (Section 3.4)	Validation result meets the threshold in Section 5.2
3 — Integration	Delivery of the outputs to each target system	Outputs received and checked by the IT team of «CUSTOMER»
4 — Parallel run and go-live	«NUMBER» scheduled runs checked with the business owner before go-live	Go-live approved per use case
5 — Hypercare and handover	Hypercare of «NUMBER» weeks, runbook, handover sessions to the data and IT teams	Runbook handed over; handover approved

Table 8 — Phases and exit criteria.

4.2 Effort and Durations

The effort consists of the MLOps foundation (once), the effort per use case by band and the effort per integration. Kick-off, parallel run, hypercare and handover are included in these lines.

Item	Effort (pd)	Duration (weeks)
MLOps foundation (once; includes Phase 0)	«NUMBER»	«NUMBER»
«FAMILY — RETENTION & CHURN» «USE CASE»	«NUMBER»	«NUMBER»
«FAMILY — DEMAND & SUPPLY» «USE CASE»	«NUMBER»	«NUMBER»
«FAMILY — PRICING & PERSONALIZATION» «USE CASE»	«NUMBER»	«NUMBER»
«FAMILY — RISK & COLLECTIONS» «USE CASE»	«NUMBER»	«NUMBER»
«FAMILY — INDUSTRIAL OPERATIONS» «USE CASE»	«NUMBER»	«NUMBER»
Integrations («NUMBER» target systems)	«NUMBER»	Within the use case phases
Total	«NUMBER»	«NUMBER» weeks

Table 9 — Effort and durations. Band «BAND».

After the MLOps foundation, the use cases can be taken into production one after the other or in parallel. The total duration depends on this order and on the team size in Section 8. «AX DOĞRULASIN»

4.3 Schedule

Phase	Scope	Planned Start	Planned End
0 – 1	Kick-off and MLOps foundation	«dd Month yyyy»	«dd Month yyyy»
2 – 4	«FAMILY — RETENTION & CHURN» «USE CASE»	«dd Month yyyy»	«dd Month yyyy»
2 – 4	«FAMILY — DEMAND & SUPPLY» «USE CASE»	«dd Month yyyy»	«dd Month yyyy»
2 – 4	«FAMILY — PRICING & PERSONALIZATION» «USE CASE»	«dd Month yyyy»	«dd Month yyyy»
2 – 4	«FAMILY — RISK & COLLECTIONS» «USE CASE»	«dd Month yyyy»	«dd Month yyyy»
2 – 4	«FAMILY — INDUSTRIAL OPERATIONS» «USE CASE»	«dd Month yyyy»	«dd Month yyyy»
5	Hypercare and handover	«dd Month yyyy»	«dd Month yyyy»

Table 10 — Schedule. Dates are finalised in Phase 0.

The schedule applies provided the assumptions in Section 7.1 hold. Delays in data access, data feeds, target system work or approvals of «CUSTOMER» move the schedule by the same amount.

5. Deliverables and Acceptance

5.1 Deliverables

Code and pipelines are delivered in the repositories and on the platform of «CUSTOMER». Documents are delivered in Turkish, in editable format together with a PDF copy.

#	Deliverable	Phase	Content
D1	Production plan	0	Confirmed scope, sizing values, platform, target systems, frequency, schedule
D2	MLOps foundation	1	Repository, CI/CD, pipeline template, model registry, scheduling, monitoring, environments
D3	Production model per use case	2	Retrained model in the registry; feature and model definitions
D4	Scheduled pipelines and output tables	2	Feature, training and scoring pipelines; output tables on «PLATFORM»
D5	Integrations	3	Delivery of the outputs to each target system in Table 4, with interface description
D6	Monitoring and report	2 – 3	Drift and performance thresholds, alerts, report of model performance and business metrics
D7	Validation and go-live report per use case	2 – 4	Results against the criteria in Section 5.2; parallel run record
D8	Value measurement report	5	Business metric of each use case against the control group for the period measured until handover «AX DOĞRULASIN»
D9	Runbook and handover	5	Operation of the pipelines, retraining, incident handling, rollback to the previous model version; handover sessions

Table 11 — Deliverables.

5.2 Acceptance

Each use case is accepted on its own. The technical criteria below are the acceptance criteria; thresholds are agreed with «CUSTOMER» in Phase 0, based on the Pilot result. «AX DOĞRULASIN»

Criterion	Measure	Threshold
Model performance	Technical metric on hold-out data, by family: AUC or Gini and lift in the top decile (Retention & Churn, Risk & Collections); WAPE / MAPE and bias against the current method on a back-test period (Demand & Supply); AUC, or precision@k and recall@k, or error of the price response model (Pricing & Personalization); prediction error or classification accuracy (Industrial Operations)	At least «NUMBER» per metric
Pipeline reliability	Consecutive scheduled runs completed without manual intervention	«NUMBER» consecutive runs
Delivery	Outputs present in each target system in the agreed format and time	All integrations in Table 4
Monitoring	Drift and performance metrics calculated on every run; alerts tested	Test record available

Table 12 — Acceptance criteria per use case.

The business metric is measured against the control group and reported: churn or segment downgrade (Retention & Churn); stock-outs, overstock and stock holding cost (Demand & Supply); conversion and revenue in an A/B test (Pricing & Personalization); transition into delinquency or approval rate at the same NPL ratio (Risk & Collections); energy consumption, deviation or yield (Industrial Operations). It is not an acceptance criterion and is not guaranteed. «AX DOĞRULASIN»

«CUSTOMER» reviews the validation and go-live report within «NUMBER» working days. If no written objection is received in that period, the use case is deemed accepted.

6. Customer Responsibilities

«CUSTOMER» is responsible for the items below. A delay in any of them moves the schedule by the same amount. «AX DOĞRULASIN»

- Appoint a project sponsor and, for each use case, a business owner (for example marketing, planning or supply chain, pricing, collections or credit risk, production or maintenance).

- Appoint a data owner and provide access to the data used by the models and to the platform at the start of Phase 0.
- Keep the data feeds used by the models running on schedule and inform Komtaş of changes in the source systems.
- Provide the IT team for the integration of the target systems and make the changes needed inside those systems.
- Ensure that the use of customer data in the models has a legal basis under KVKK and the internal policies of «CUSTOMER».
- Act on the outputs (campaigns, replenishment, prices, collection actions, set points) and keep the agreed control group so that business value can be measured.
- Approve go-live of each use case and take part in the handover sessions.

7. Assumptions and Out of Scope

7.1 Assumptions

The duration, effort and fee in this proposal are based on the assumptions below. If they do not hold, scope, schedule and fee are reassessed under change management. «AX DOĞRULASIN»

- The values in Table 2 are taken from «PILOT REF» and the proposal input form. If, when confirmed in Phase 0, a value moves the band up, effort and fee are updated under change management.
- The data used in the Pilot is on «PLATFORM» or is loaded to it by existing pipelines of «CUSTOMER».
- The features used in the Pilot are available in production with the same definition and at the agreed frequency.
- Batch scoring at the agreed frequency is sufficient; real-time (online) scoring is not required.
- Cloud consumption and platform licence costs fall under the account and commitment of «CUSTOMER».
- The work starts within «NUMBER» working days of contract signature and is carried out mainly remotely; meetings can be held on site in «CITY».
- The working language and the language of the deliverables is Turkish.

7.2 Out of Scope

The topics below are outside the scope of this proposal. If requested, they are assessed and proposed separately. «AX DOĞRULASIN»

- Migration of the data platform (Section 10).
- Cleansing of data in the source systems.
- Execution of campaigns and other business actions based on the outputs.
- A guaranteed business result.
- Real-time (online) scoring, unless specified in Section 2.4.
- An enterprise MLOps or AI Factory platform beyond the MLOps foundation in Section 3.3.
- Migration of existing models or data flows to another analytics platform.
- Generative AI and RAG applications.
- Use cases not listed in Section 1.
- Changes inside the target systems (Section 2.5).
- Monitoring, retraining and reporting after hypercare; offered as service C Run (Section 10).

7.3 Risks and Change Management

Risk	Impact	Mitigation
Data has changed since the Pilot	Model performance	Validation on current data in Phase 2; retraining before go-live
A Pilot feature is not available in production	Effort, model performance	Features checked in Phase 0; replacement agreed with the data owner
Target definition changes (churn, delinquency, quality defect)	Scope	Definitions from the Pilot fixed in writing in Phase 0; a change is handled as a change request
«FAMILY — INDUSTRIAL OPERATIONS» Gaps in sensor or laboratory data; change in operating conditions	Model performance	Data gaps checked in Phase 0; operating limits agreed with the process engineers; retraining when conditions change
The business does not act on the outputs	Business value not measurable	Business owner and control group named in Phase 0; use of the outputs followed in the governance meetings
Target system work delayed	Schedule	Interface agreed in Phase 0; outputs kept in output tables on «PLATFORM» until the target system is ready

Table 13 — Risks.

Any request affecting scope, schedule or fee is handled through a written change request. The request is submitted with its impact and fee and is not implemented without approval.

8. Team

8.1 Komtaş Team

The work is led by a Data Scientist. «AX DOĞRULASIN»

Role	Responsibility	Involvement
Data Scientist (lead)	Models, features, validation, value measurement, technical decisions and quality	All phases
ML Engineer	MLOps foundation, pipelines, model registry, scheduling, monitoring	All phases; main role in Phases 1–2
Data Engineer	Data feeds, feature tables, output tables, integration with the target systems	Phases 1–3
Business Analyst / Engagement Lead	Plan, business owners, value measurement design, change requests, reporting	Part time, all phases

Table 14 — Komtaş team and roles.

- All team members assigned to the work are employees of Komtaş; no subcontractors are used. «AX DOĞRULASIN»
- If a change in the team is needed, the replacement has at least equivalent qualifications and is notified to «CUSTOMER» in advance.

8.2 Participation Expected from «CUSTOMER»

Role	Expected Contribution	Involvement
Project Sponsor	Decisions, priorities, phase approvals	Phase transitions
Business Owner per use case	Use of the outputs, control group, go-live approval «AX DOĞRULASIN»	Phases 0, 4 and 5
Data Owner	Access, data feeds, data definitions	Phases 0–2
«FAMILY — INDUSTRIAL OPERATIONS» Process Engineers	Operating limits, set points, laboratory and sensor data definitions	Phases 0–4
IT / Integration Team	Interfaces and changes in the target systems, platform access	Phases 1 and 3

Table 15 — Expected customer-side participation. Total estimated customer effort is «NUMBER» person-days.

8.3 Governance and Reporting

- Weekly progress meeting: status per use case, open items and risks.

- Monthly steering committee: schedule, use of the outputs, value measurement and change requests.
- Written approval at phase exits and at the go-live of each use case.

9. Effort and Fees

> *FILL NOTE — the fees in this section are entered by sales. Effort comes from the proposal input form (predictive-analytics-3b-production-girdi-formu): use cases × band effort + MLOps foundation once + integrations. Draft values:*
>
> *Effort per use case: S ~40 pd, M ~60 pd, L ~90 pd. MLOps foundation, once: ~20–30 pd.* «AX DOĞRULASIN»
>
> *Integration: «NUMBER» pd per target system; there is no draft value yet. The values are not based on recorded effort and are updated with the data of the Analytics team. Delete this box before sending.*

9.1 Basis of the Fee

The fee consists of a fixed fee for the MLOps foundation, counted once, a fixed fee per use case in scope and a fixed fee per integration with a target system. Each line covers the scope described in Sections 2 and 3 for that line.

9.2 Fees

#	Fee Line	Effort (pd)	Fee (excl. VAT)
F1	MLOps foundation (once, used by all use cases)	«NUMBER»	«fee»
F2	«FAMILY — RETENTION & CHURN» «USE CASE» in production	«NUMBER»	«fee»
F3	«FAMILY — DEMAND & SUPPLY» «USE CASE» in production	«NUMBER»	«fee»
F4	«FAMILY — PRICING & PERSONALIZATION» «USE CASE» in production	«NUMBER»	«fee»
F5	«FAMILY — RISK & COLLECTIONS» «USE CASE» in production	«NUMBER»	«fee»
F6	«FAMILY — INDUSTRIAL OPERATIONS» «USE CASE» in production	«NUMBER»	«fee»
I1	Integration — «TARGET SYSTEM»	«NUMBER»	«fee»
I2	Integration — «TARGET SYSTEM»	«NUMBER»	«fee»
	Total	«NUMBER»	«fee»

Table 16 — Fees per line. Band «BAND»; one row per use case and one row per integration in Table 4.

A use case added later uses the MLOps foundation built in this proposal; only the fee of the use case and of its integrations applies.

9.3 Payment Schedule

Payment Item	Rate / Basis	Amount (excl. VAT)
Phase 1 exit approval — MLOps foundation	F1	«amount»
Go-live approval of each use case	«%» of the use case fee	«amount»
Acceptance of each integration	Integration fee	«amount»
Handover approval of each use case	«%» of the use case fee	«amount»

Table 17 — Payment schedule.

9.4 Change Requests

Work outside the scope of this proposal is offered on the basis of person-days × day rate («day rate», excl. VAT) and is carried out only after written approval.

9.5 Not Included in the Fee

- Cloud consumption and licence costs of «PLATFORM» (for example BigQuery and Vertex AI on Google Cloud, Dataiku licences, Azure Machine Learning).
- Licences and changes of the target systems.

- Monitoring, retraining and reporting after hypercare (service C Run).
- Additional work approved under change requests.

10. Next Step

After go-live, the models can be operated and extended step by step. The usual follow-on work is:

Need	Follow-On Work
Models stay accurate after go-live	C Run: monthly monitoring, retraining and reporting
Further business questions	Next use case from the catalogue on the same MLOps foundation
Data in on-premises ETL or warehouse	Data platform modernisation
Many models and teams	MLOps / AI Factory platform; Analytic CoE Enablement Program
Governance of models	AI Governance Readiness

Table 18 — Follow-on work.

10.1 Run

After hypercare, the models can be operated by Komtaş under service C Run: drift and performance monitoring, scheduled retraining, a monthly report and small improvements from a monthly capacity. Run is offered in a separate proposal. «AX DOĞRULASIN»

10.2 Next Use Case

All use cases in the catalogue (Annex 1) use the same MLOps foundation. A further use case can be started with a Pilot or, if a validated model exists, directly with Production.

10.3 Data Platform

Where the data used by the models sits in on-premises ETL tools or warehouses, the move of the data platform to the cloud can be planned together with the models.

10.4 Analytics Organisation and AI Governance

Where more models and teams follow, an MLOps / AI Factory platform and the Analytic CoE Enablement Program build the platform, team and roadmap. AI Governance Readiness reviews how models are approved, documented and monitored across the organisation.

10.5 Consumption

Komtaş is a Google Cloud Premier partner and a Microsoft Azure partner. Where «PLATFORM» is a cloud platform, the cloud consumption of «CUSTOMER» can be purchased through Komtaş; the commercial terms are set out in a separate proposal.

11. Commercial Terms

11.1 General Terms

- All fees in this proposal are stated in Turkish lira and exclude VAT. Statutory taxes, duties and charges are not included in the fees.
- The proposal is valid until «proposal date + 1 month».
- The fees are valid on condition that the work starts by «price validity date». If the start moves beyond that date for any reason, the fees and pricing terms are reassessed by mutual agreement in the light of the economic and financial conditions at the time of the work.
- The proposal is based on the Pilot «PILOT REF» and the assumptions in Section 7.1; if the basis changes materially, the proposal is renewed.
- The fee is calculated per line as set out in Section 9: MLOps foundation once, per use case and per integration.
- Cloud consumption and platform licences are not part of this proposal.

- Payment follows the payment schedule in Section 9. Invoices are payable within «NUMBER» days of the invoice date.
- Scope changes are managed through a written change request; no change is implemented before its impact analysis is approved.
- The parties keep confidential all information they access within the scope of the work. A separate non-disclosure agreement is signed where needed.
- The rights of use of the deliverables, including the models, pipelines and code, belong to «CUSTOMER»; Komtaş retains its intellectual property rights in its methodology, templates and general know-how.
- This proposal forms an integral annex to the contract to be signed by the parties. In the event of a conflict between the contract and the proposal, the contract prevails.
- In the event of force majeure, the parties inform each other in writing and the impact on the schedule is reassessed by mutual agreement.

11.2 Proposal Contact Details

Questions on this proposal and its scope can be addressed to the contacts below.

Area of Responsibility	Full Name	Title	Contact
Commercial	«full name»	«title»	«e-mail / phone»
Technical	«full name»	«title»	«e-mail / phone»

11.3 Proposal Annexes

This proposal is submitted with the following annexes:

- ANNEX-1 — Use-Case Family Pages
- ANNEX-2 — About Komtaş

ANNEX-1 — Use-Case Family Pages

Predictive Analytics groups its use cases into five families. This annex describes each family in scope: the use cases, the business problem, the data needed, the output and the success metrics. Measured results of earlier projects are in Table 25 (Annex 2).

> *FILL NOTE — keep only the pages of the families in scope and delete the family marker in their headings.*

«FAMILY — RETENTION & CHURN» Retention & Churn

Item	Description
Use cases	Path to Churn; "balance" churners (incoming funds withdrawn soon after arrival); asset churn and segment downgrade; subscription churn; customer satisfaction index «AX DOĞRULASIN»
Typical sectors	Banking, fitness and subscription businesses, telecommunications, retail loyalty programmes, insurance
Business problem	Which customers are likely to leave, cancel or move to a lower segment, which events lead there, and where retention actions are focused.
Data needed	At least 12 to 24 months of customer, product or membership, transaction, usage or visit, channel interaction, campaign and complaint data; churn or downgrade definition agreed with the business owner
Output	Probability and main drivers per customer; critical events for alert and action scenarios; refreshed «FREQUENCY»
Technical success metric	AUC or Gini and lift in the top decile on hold-out data
Business metric	Churn or segment downgrade in the test group against the control group
Field references	Asset churn prediction; subscription churn (Table 25)

Table 19 — Retention & Churn family.

«FAMILY — DEMAND & SUPPLY» Demand & Supply

Item	Description
Use cases	Demand forecasting (retail, manufacturing); inventory productivity analysis; visitor and sales prediction (shopping malls); schedule optimization «AX DOĞRULASIN»
Typical sectors	Retail and e-commerce, manufacturing, shopping malls, FMCG and distribution
Business problem	How much will be sold, produced or how many visitors will come per product, store, site or channel in the coming periods; planning, replenishment and stock use the forecast to reduce stock-outs and overstock.
Data needed	At least 2 to 3 years of sales at product, store and day or week level; price and promotion; stock-outs; calendar and holidays; store or site attributes; external data where relevant (weather, inflation, card spending, market data)
Output	Forecast per series for the horizon «HORIZON», refreshed «FREQUENCY»; scenario runs where needed
Technical success metric	WAPE / MAPE and bias against the current method of «CUSTOMER» on a back-test period
Business metric	Stock-outs, overstock and stock holding cost
Field references	Demand prediction; visitor and sales prediction; demand forecasting in retail (Table 25)

Table 20 — Demand & Supply family.

«FAMILY — PRICING & PERSONALIZATION» Pricing & Personalization

Item	Description
Use cases	Next best offer; product bundling; recommendation; customer lifetime value; customer segmentation; customer contact policy; onboarding analytics; deposit pricing; credit pricing; list price and markdown optimization; dynamic pricing «AX DOĞRULASIN»
Typical sectors	Banking, retail and e-commerce, spare parts and distribution, media
Business problem	Which offer, product, price or contact to give to each customer or segment, and when; which customers bring the most value over time.
Data needed	At least 6 to 24 months of customer, transaction and interaction data (views, baskets, purchases, offers and responses); product catalogue and hierarchy; price history and negotiation data for pricing
Output	Offer or top-N products per customer or context; segment or lifetime value per customer; list prices and markdown rules per product group and customer segment; refreshed «FREQUENCY»
Technical success metric	AUC for propensity models; precision@k and recall@k for recommendation; error of the price elasticity model on hold-out data for pricing
Business metric	Conversion, click-through and revenue in an A/B test against a control group
Field references	List price and markdown optimization (Table 25). Recommendation has no field reference yet; it is built with the same lifecycle and MLOps foundation.

Table 21 — Pricing & Personalization family.

«FAMILY — RISK & COLLECTIONS» Risk & Collections

Item	Description
Use cases	Collection analytics and optimization; credit risk model enhancement, including graph analytics on relational data; dynamic client limit; risk analytics «AX DOĞRULASIN»
Typical sectors	Banking and financial services
Business problem	Which customers are likely to enter delinquency or default in the next period, and how contact policy, credit approval and limits are differentiated by risk.
Data needed	Credit, repayment and delinquency history; customer and product data; payments, transfers and relationships between customers (partnerships, guarantees); contact history of the collection channels
Output	Delinquency or default probability and main drivers per customer; contact policy by risk level; input to credit and limit decisions; refreshed «FREQUENCY»
Technical success metric	Gini or AUC on hold-out data; feature importance and direction of impact reported to decision makers
Business metric	Transition into the first delinquency cycle, or credit approval rate at the same NPL ratio, against a control group
Field references	Collection optimization; credit risk model enhancement (Table 25)

Table 22 — Risk & Collections family.

«FAMILY — INDUSTRIAL OPERATIONS» Industrial Operations

Item	Description
Use cases	Energy optimization (HVAC, furnace); predictive quality and quality classification; production yield optimization; predictive maintenance; process unit prediction (refinery); fixed asset utilization «AX DOĞRULASIN»
Typical sectors	Manufacturing (for example glass, pharmaceuticals), refining and petrochemicals
Business problem	How process values (energy use, temperature, quality, yield, equipment condition) will develop, and which set points keep them within the operating limits at the lowest energy or cost.
Data needed	Sensor and IoT data; laboratory samples; production and quality records; set points and operating limits; maintenance records; ambient conditions
Output	Predicted process values for the coming hours; recommended set points within the operating limits; defect class or quality score; refreshed «FREQUENCY»
Technical success metric	Prediction error on hold-out data (for example MAE or MAPE); classification accuracy for quality
Business metric	Energy consumption, deviation, yield or labour time in test units or periods against a control group
Field references	Energy optimization in HVAC; furnace energy, furnace temperature control, glass quality classification and production yield; process unit prediction (Table 25)

Table 23 — Industrial Operations family.

ANNEX-2 — About Komtaş

> *FILL NOTE — the figures in Table 24 and in the bullets below come from the proposals of 2026. Check them against the current corporate profile before sending.*

Komtaş is an independent information technology company that has specialised in enterprise data management in Türkiye since 1989. More than 35 years of focus on a single field has given it an end-to-end view that follows data from the moment it is created to the moment a decision is taken.

Indicator	Value
Founded	1989 — more than 35 years of continuous data-focused activity
Specialist staff	250+ data engineers, architects, consultants and analysts
Offices	Istanbul, Ankara, Izmir, Dubai
Strategic partnerships	10+ global technology vendors
Focus area	Data governance, data architecture, analytics, artificial intelligence and AI agent based solutions
Positioning	"From Data to Agents" — integrated capability from data to AI agents

Table 24 — Komtaş corporate profile.

Service Capabilities

The service portfolio of Komtaş covers the whole life cycle of a data initiative:

- **Data Governance and Data Management:** data governance maturity assessment, framework and organisation design, policy and process design, data catalogue, business glossary, data quality management, master data management, data privacy and classification.
- **Data Architecture and Engineering:** data warehouse and data lake architectures, lakehouse design, data integration and streaming architectures, cloud and on-premises modernisation.
- **Analytics and Artificial Intelligence:** enterprise reporting, advanced analytics, machine learning models, generative AI applications and AI governance.
- **Managed Services and Support:** P1 / P2 / P3 support for production systems, capacity and performance management, continuous improvement services.
- **Training and Capability Building:** product training, role-based data governance training, train-the-trainer programmes.

Platforms and Cloud Partnerships

Komtaş is a Google Cloud Premier partner and has completed its partnership with Microsoft Azure. Models are delivered on Google Cloud, Dataiku, Azure Machine Learning or open-source stacks; data platform, models, operation and cloud consumption can be provided by the same organisation.

References and Measured Results

FILL NOTE — the reference names and results in Table 25 come from the references deck of the Komtaş Analytics team (2026). Rows may be limited to the families in scope. Do not add names, contact persons or result figures without approval; «RESULT» stays until the customer concerned approves a figure.

Predictive analytics projects of Komtaş with measured results are listed below. Results were measured in the test group against a control group unless stated otherwise. Reference calls can be arranged on request. «AX DOĞRULASIN»

Use Case	Family	Reference	Result
Asset churn prediction	Retention & Churn	Odeabank	11% lower segment downgrade in the test group than in the control group
Subscription churn	Retention & Churn	A fitness and subscription company «GÜNER DOĞRULASIN»	«RESULT»
Demand prediction	Demand & Supply	A tyre manufacturer	7% lower stock holding cost for 7 tyre types
Visitor and sales prediction	Demand & Supply	A shopping mall management company	Forecasts for 350+ stores in 2 malls and 1.6 million visitors (no control group)
Demand forecasting (retail)	Demand & Supply	A retail company «GÜNER DOĞRULASIN»	«RESULT»
List price and markdown optimization	Pricing & Personalization	A spare-parts seller	+4% revenue at the same sales conversion in the test group; price elasticity for 30,000+ SKUs; 12 A/B test scenarios
Collection optimization	Risk & Collections	Albaraka Banking Group, TEB	3–7% reduction in transition into the first delinquency cycle; Gini 0.86; explainable AI; contact policy optimised
Credit risk model enhancement	Risk & Collections	QNB Finansbank	+4.5% credit approval rate at the same NPL ratio; Gini 0.78–0.81; graph analytics
Energy optimization in HVAC	Industrial Operations	Eczacıbaşı Monrol	2.3% lower energy consumption in the test units than in the control group
Furnace energy, furnace temperature control, glass quality classification, production yield	Industrial Operations	Şişecam	«RESULT»
Process unit prediction (continuous catalytic reformer)	Industrial Operations	An integrated oil and petrochemical group	Qualitative: more control over the process and more stable operation

Table 25 — Predictive analytics references of Komtaş and measured results.

MLOps Foundation Experience

The MLOps foundation in Section 3.3 builds on the following work: «AX DOĞRULASIN»

Organisation	Work and Platform
Şişecam	MLOps for an industrial IoT platform: model training and deployment with Google Cloud Composer and Dataiku; monitoring with conditional and periodic retraining and data and model stability checks in production on Dataiku
A bank in Azerbaijan	AI Factory: end-to-end MLOps platform built with open-source technologies, from data integration to serving the results to business units; recognised by IDC
A state bank in Türkiye	Revamp of segmentation and forecasting models with MLOps methods; data preparation, feature engineering and model evaluation standards adopted by the bank
A private bank in Türkiye	Standardisation of model development and management on Dataiku; in-database data preparation and containerised model training

Table 26 — MLOps foundation experience of Komtaş.

Why Komtaş

- **Focus:** Komtaş has worked only in the data field for more than 35 years, with more than 250 specialists and 4 offices.
- **From Pilot to Run:** Pilot, Production, Run and cloud consumption are provided by the same organisation (Section 10).